

The background features three vertical stripes on the left: a wide pink stripe, a medium blue stripe, and a narrow light beige stripe. The rest of the background is a light beige color with a pattern of small, faint pink dots arranged in a grid-like fashion, with some dots missing to create a sparse effect.

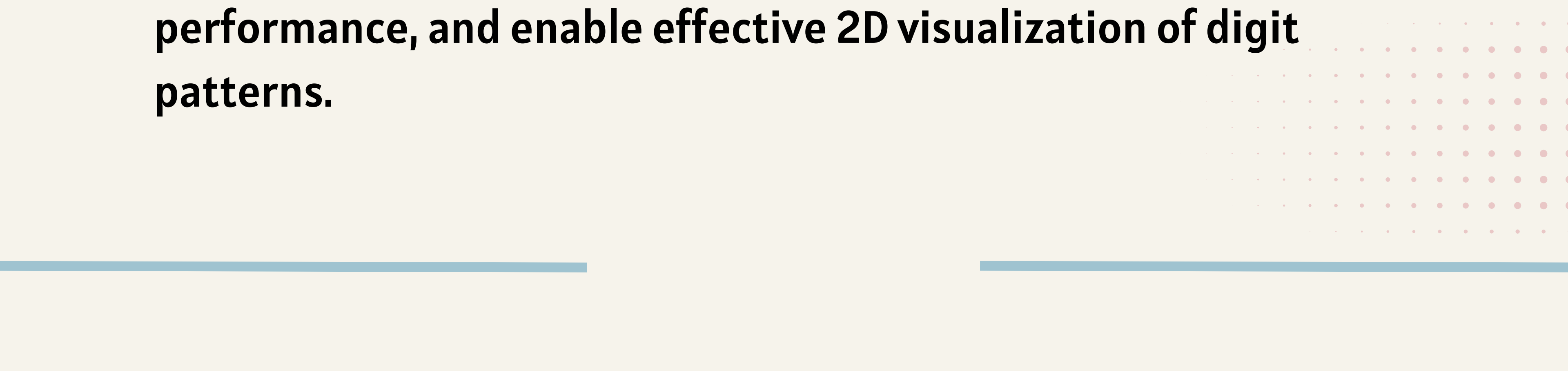
UNDERSTANDING HANDWRITTEN DIGITS PATTERNS THROUGH CLUSTERING



OBJECTIVE

The project aims to cluster handwritten digit images from the MNIST dataset using unsupervised learning techniques.

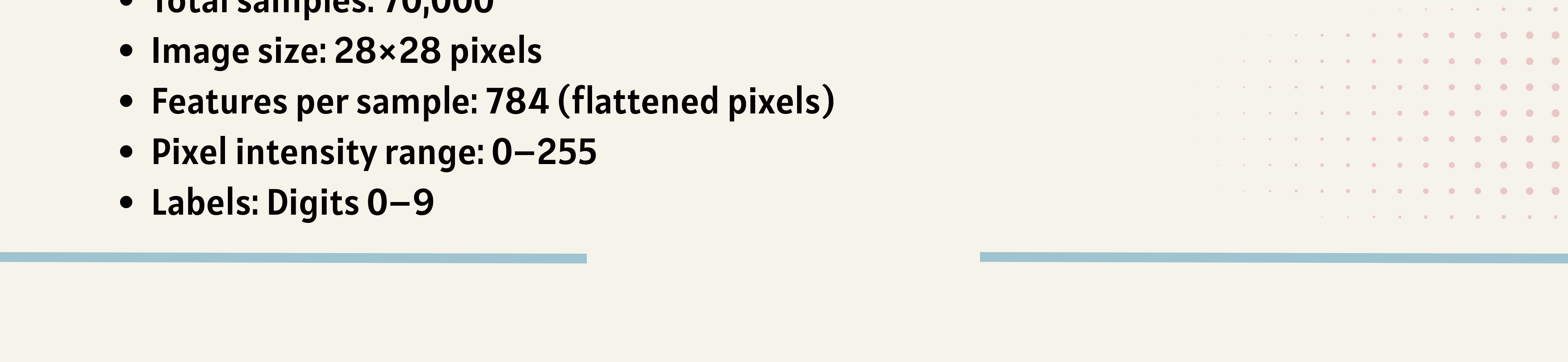
Dimensionality reduction methods like PCA, t-SNE, and SVD are applied to simplify high-dimensional data, enhance clustering performance, and enable effective 2D visualization of digit patterns.



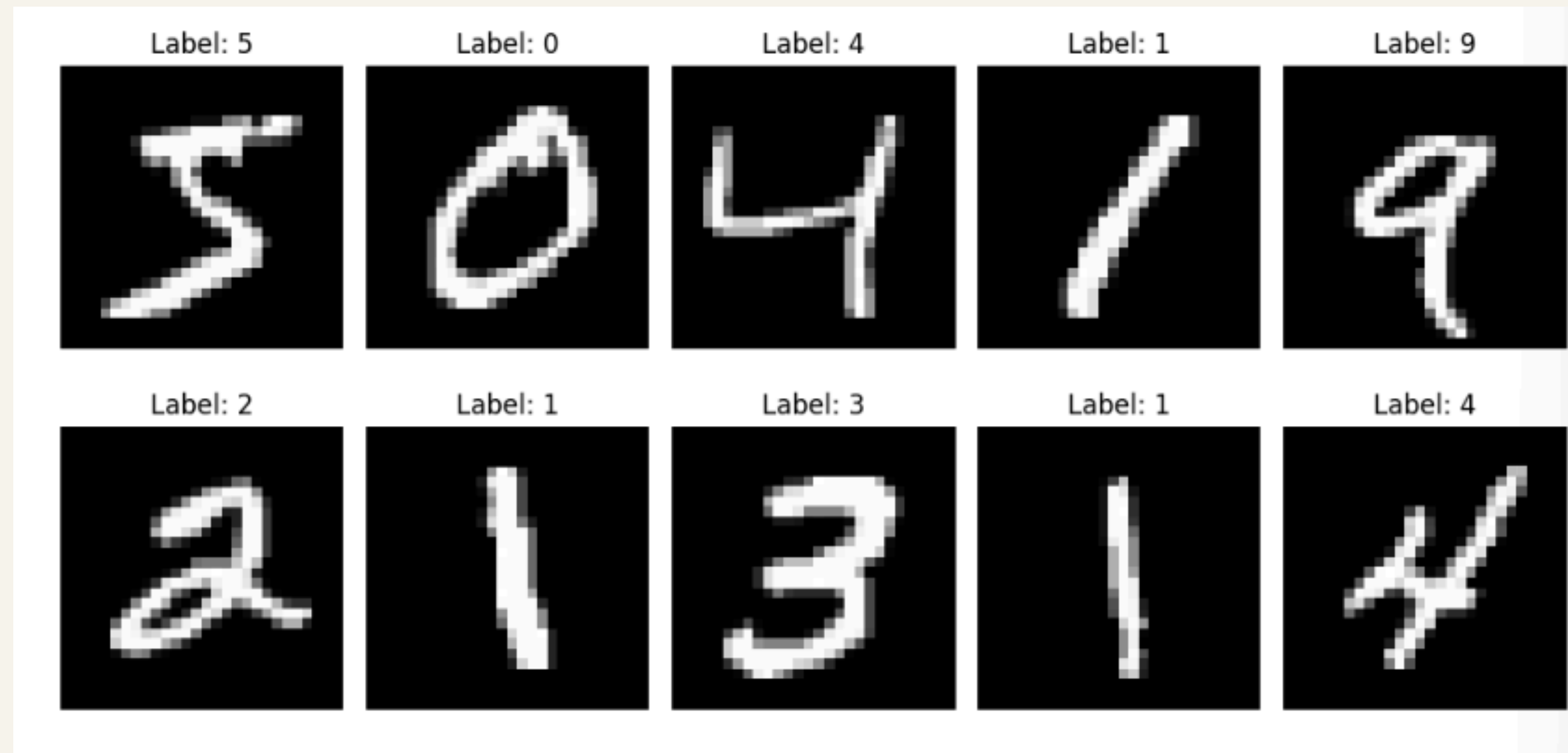


DATASET OVERVIEW

The MNIST (Modified National Institute of Standards and Technology) dataset is a widely used benchmark in machine learning and computer vision. It consists of grayscale images of handwritten digits (0–9).

- **Total samples: 70,000**
 - **Image size: 28×28 pixels**
 - **Features per sample: 784 (flattened pixels)**
 - **Pixel intensity range: 0–255**
 - **Labels: Digits 0–9**
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sample handwritten digits



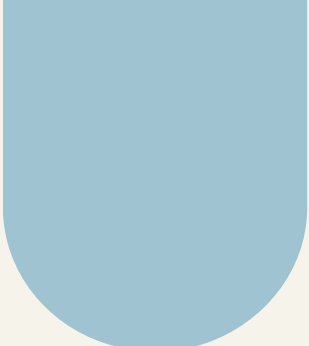
DATA PREPROCESSING

1. Feature Scaling

- Pixel values (0–255) scaled to the range -1 to 1 using StandardScaler
- Helps algorithms converge faster and improves clustering performance

2. Stratified Sampling

- Used `train_test_split` with stratification to create balanced subsets
- Applied for computationally expensive algorithms:
 - Agglomerative Clustering
 - DBSCAN
- Ensures representative distribution of all digits in the subset



DIMENSIONALITY REDUCTION

Technique Used:

- **Principal Component Analysis (PCA)**

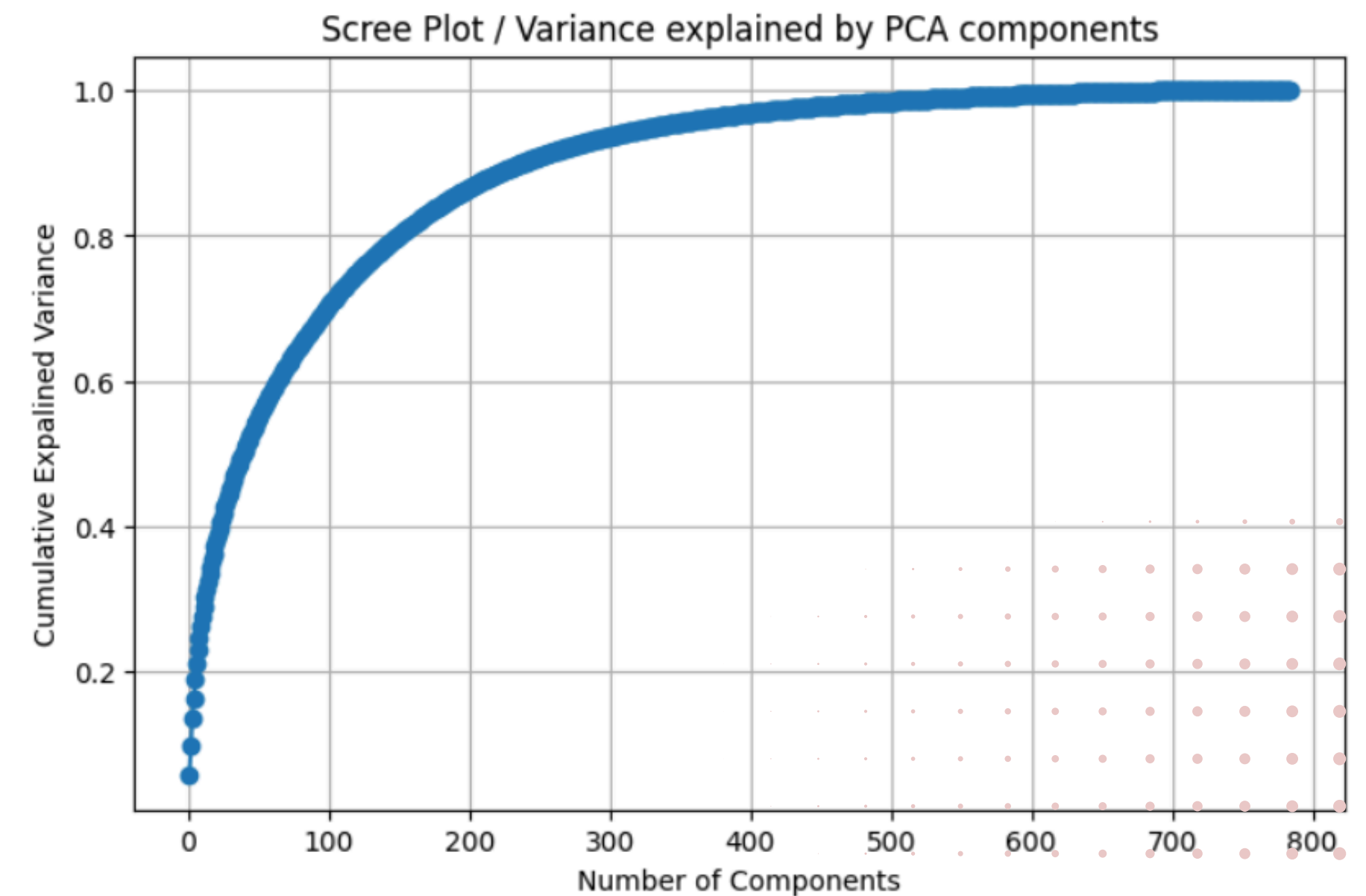
Purpose:

- **Reduce the high-dimensional 784-feature space to a lower-dimensional representation**
 - **Improve clustering performance and reduce computational cost**
 - **Enable visualization of data in 2D/3D**
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Implementation Details:

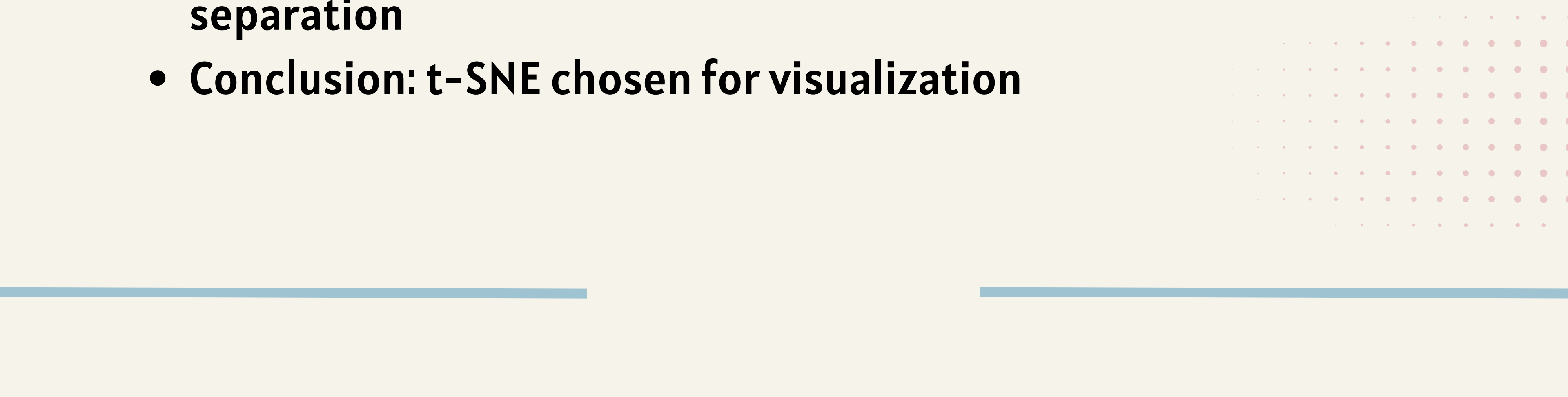
PCA applied before clustering for K-Means, Agglomerative, and DBSCAN

- We need almost 200 principal components for covering 90% variance explained
- For clustering purpose we use first 50 principal components which will cover 60% of variance

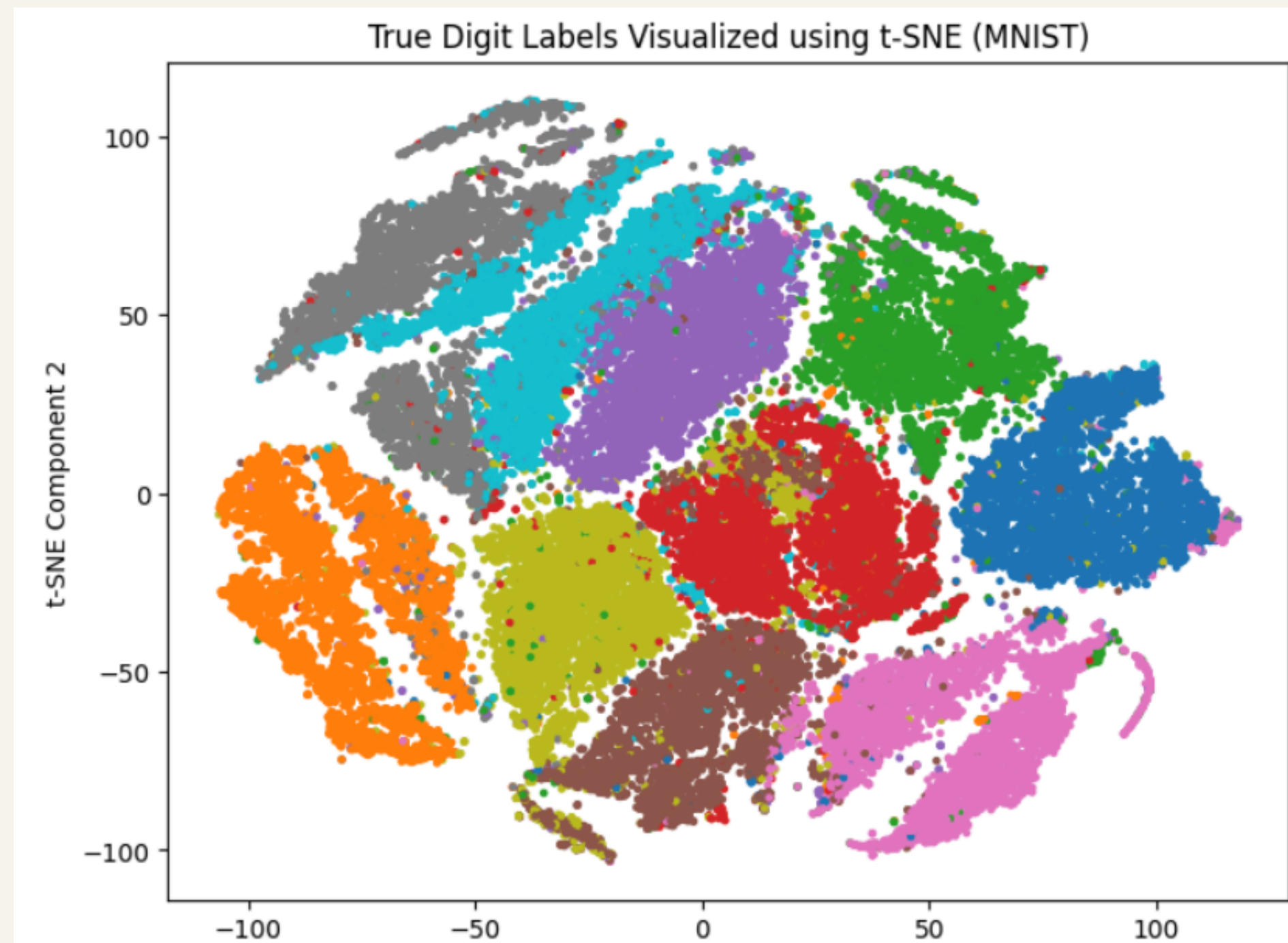
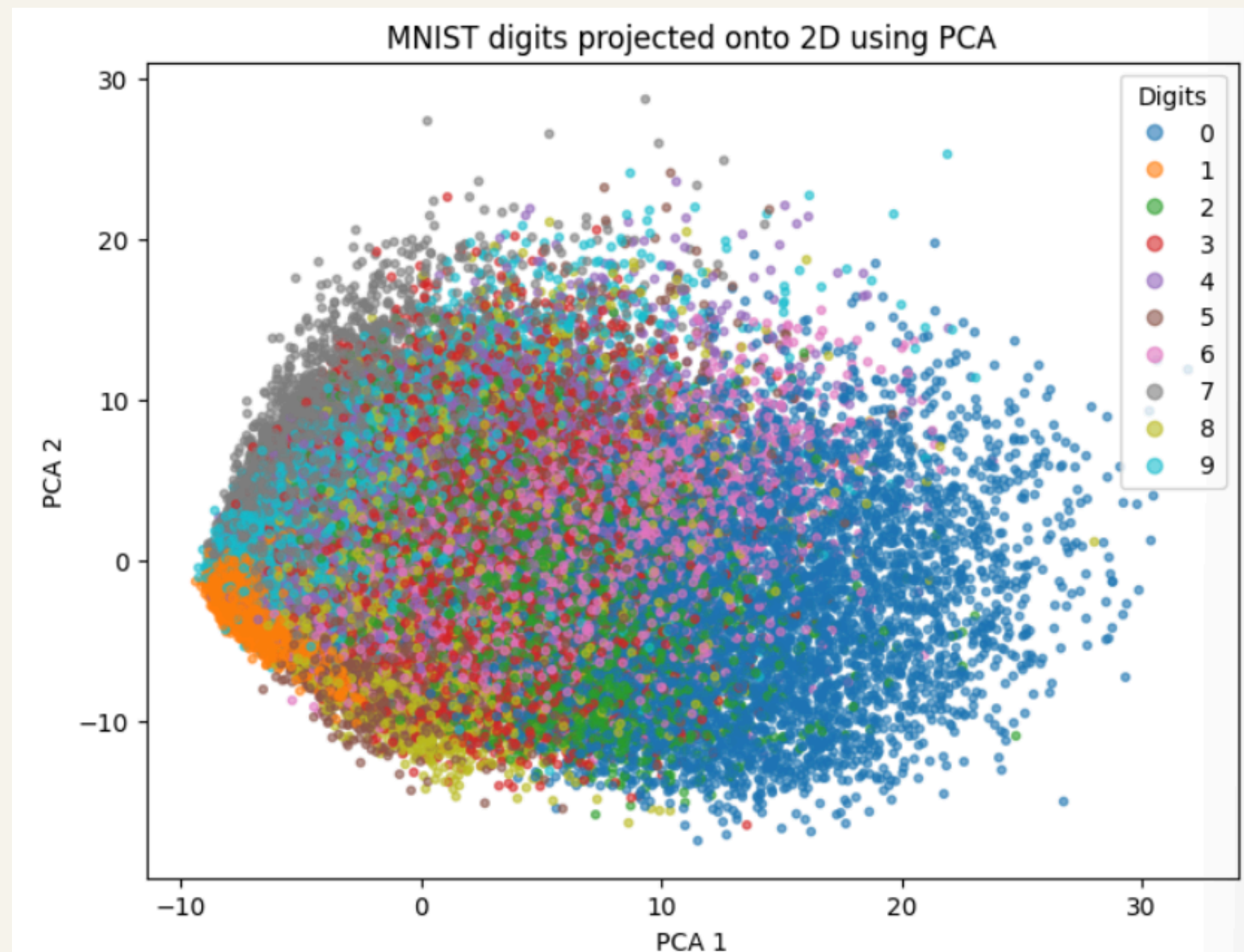




PCA vs t-SNE Visualization

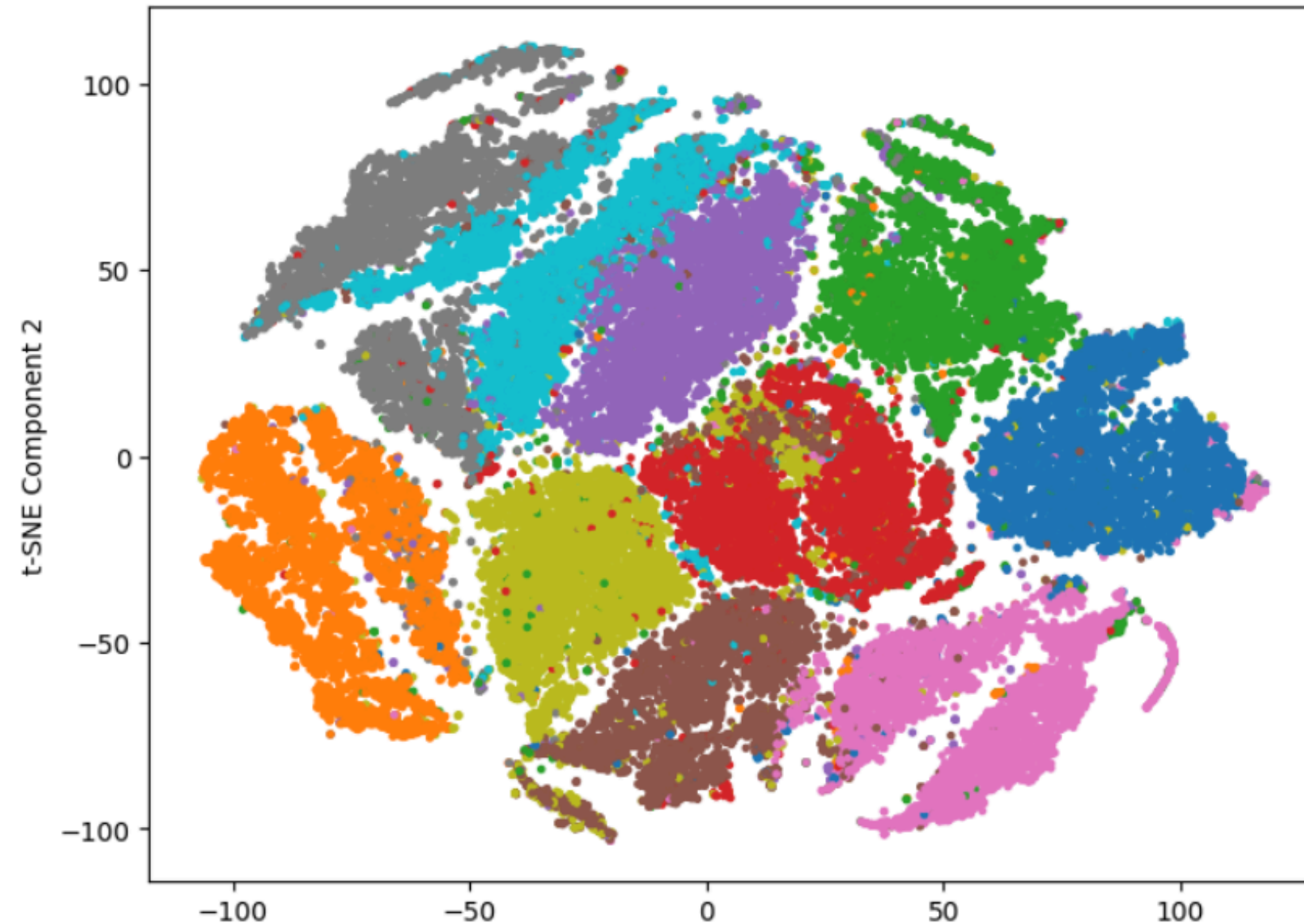
- Reduced MNIST dataset to 2D using PCA and t-SNE
 - PCA: Captures linear relationships; fails to separate complex digit patterns clearly
 - t-SNE: Preserves non-linear structures; shows better cluster separation
 - Conclusion: t-SNE chosen for visualization
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PCA vs t-SNE Visualization

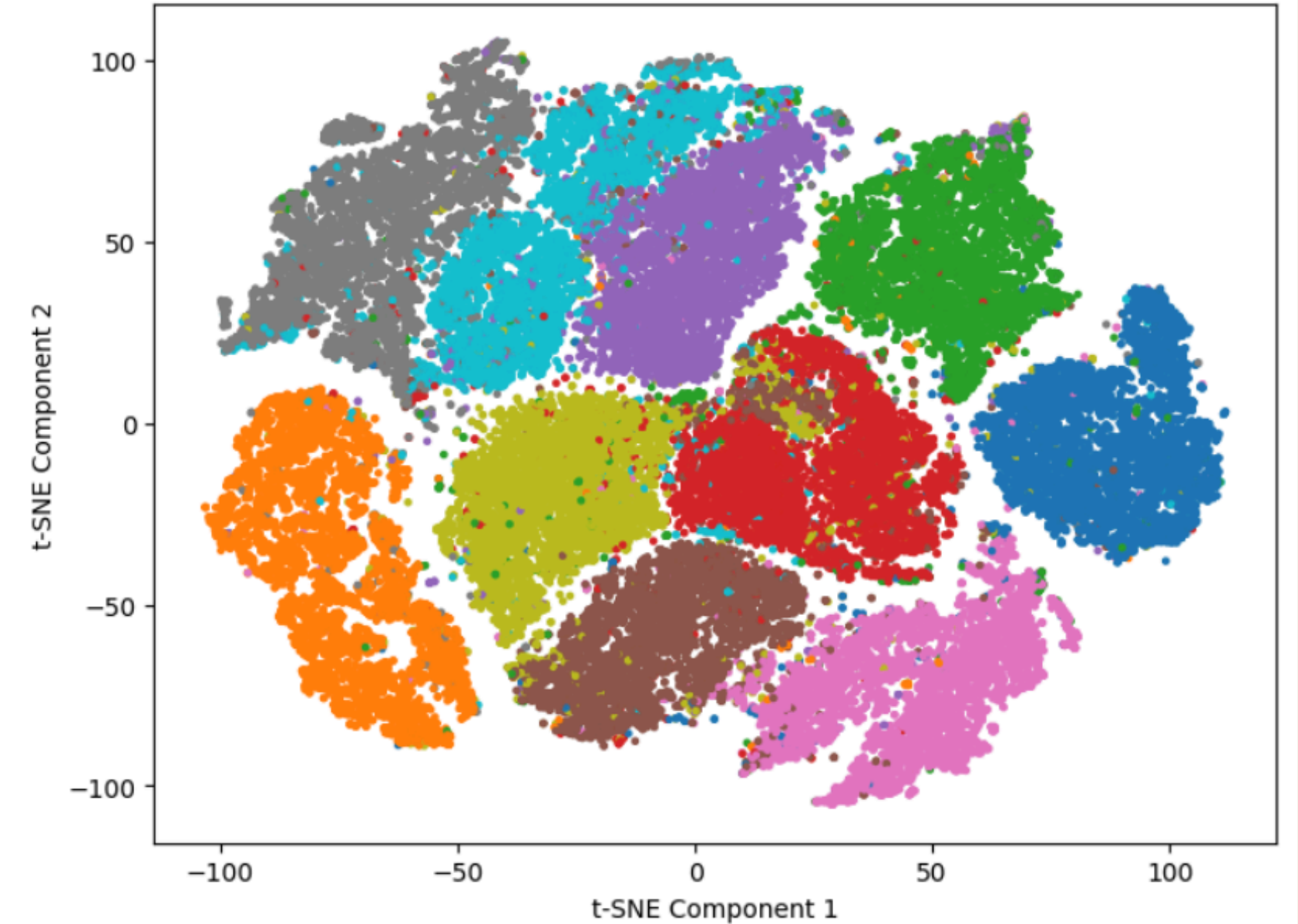


PCA 240 dimesnions vs PCA 50 dimensions

True Digit Labels Visualized using t-SNE (MNIST)

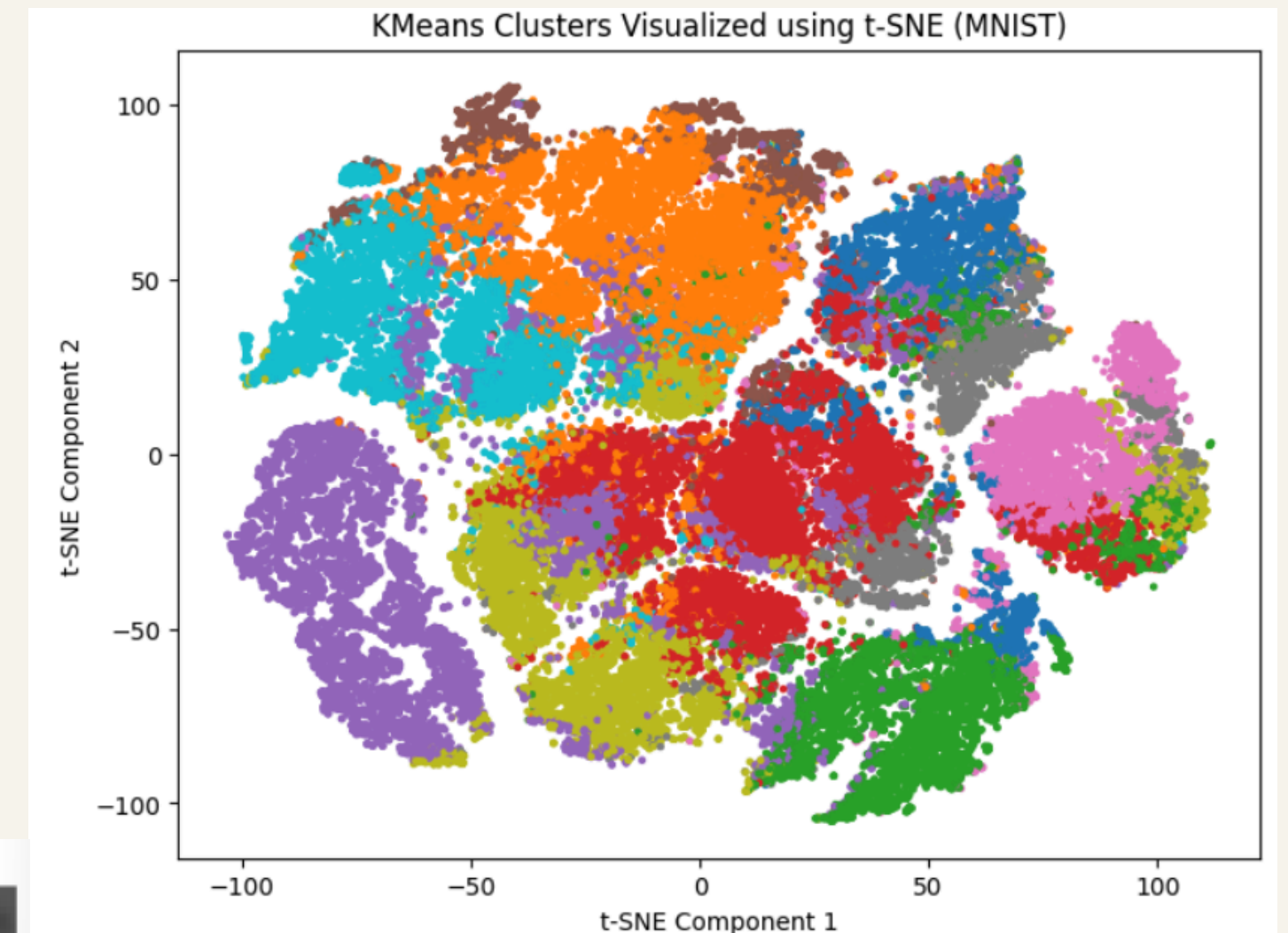
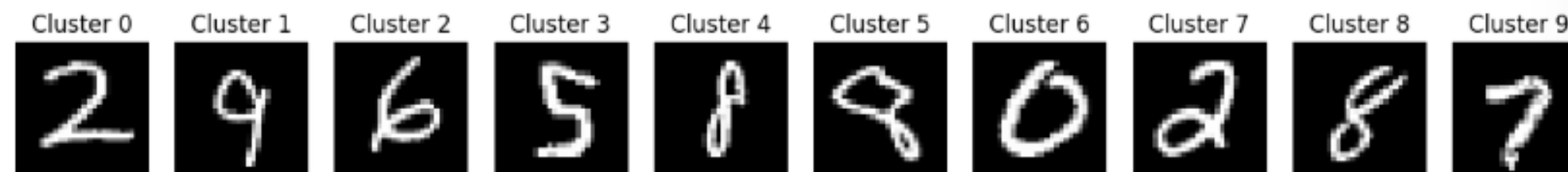
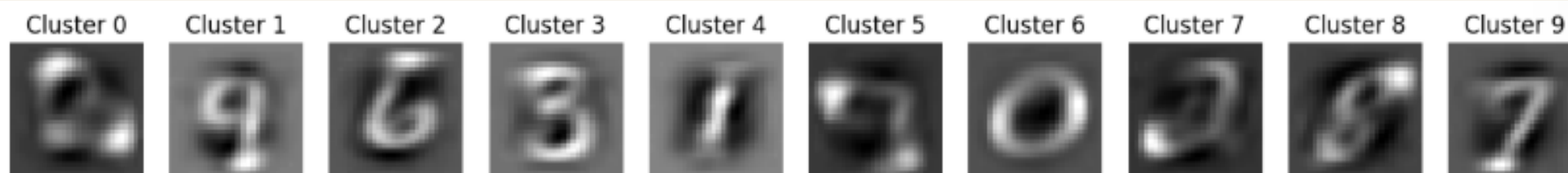


True Digit Labels Visualized using t-SNE (MNIST)



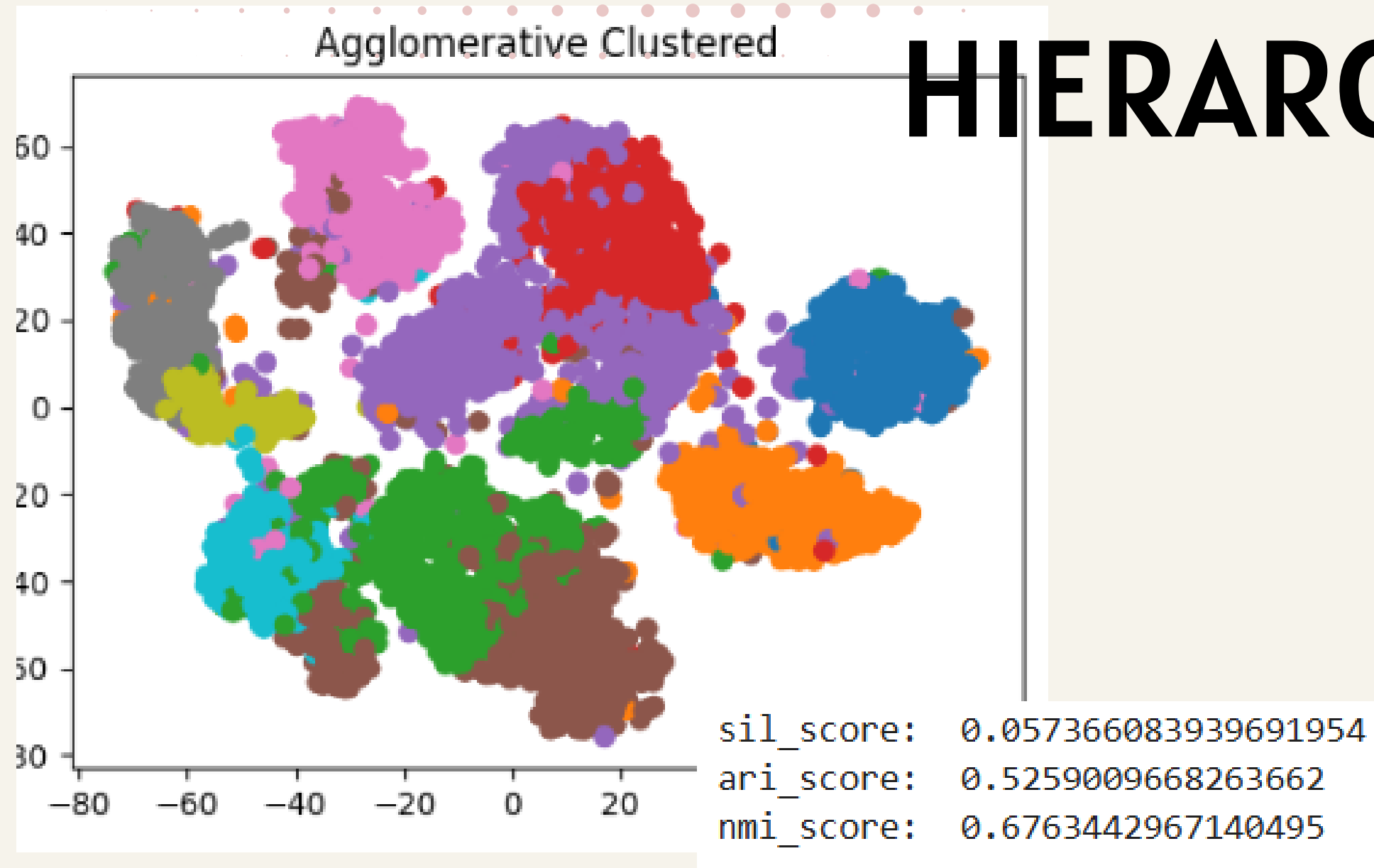
K-MEANS

- K-Means captures general digit clusters, but some overlap exists
- Performs reasonably well on MNIST, but perfect separation of all 10 digits is not achieved due to complex patterns and non-linear relationships in the dataset

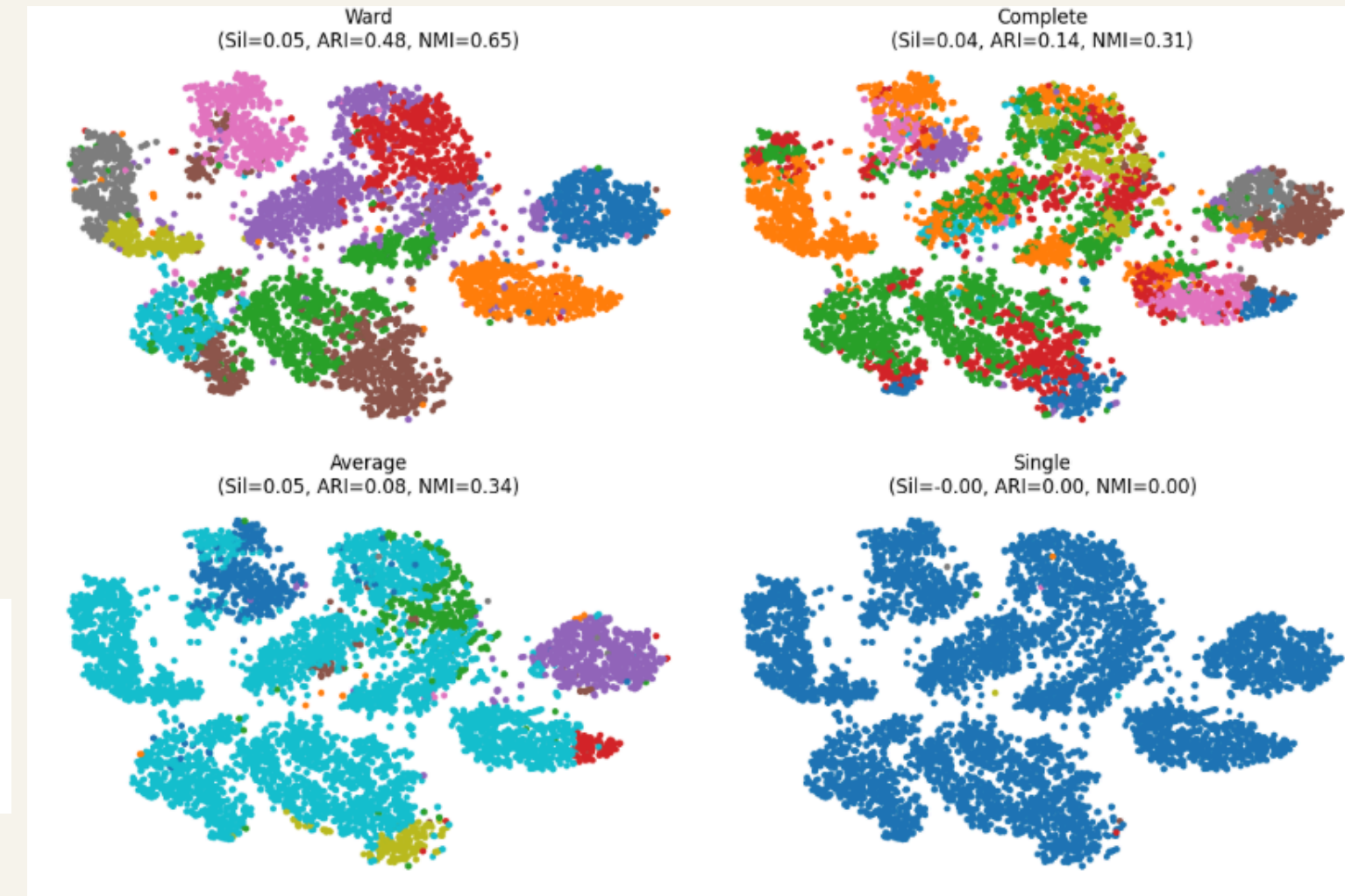


```
sil_score: 0.08170195
ari_score: 0.40759041938210666
nmi_score: 0.5207706555417181
```

HIERARCHIAL CLUSTERING

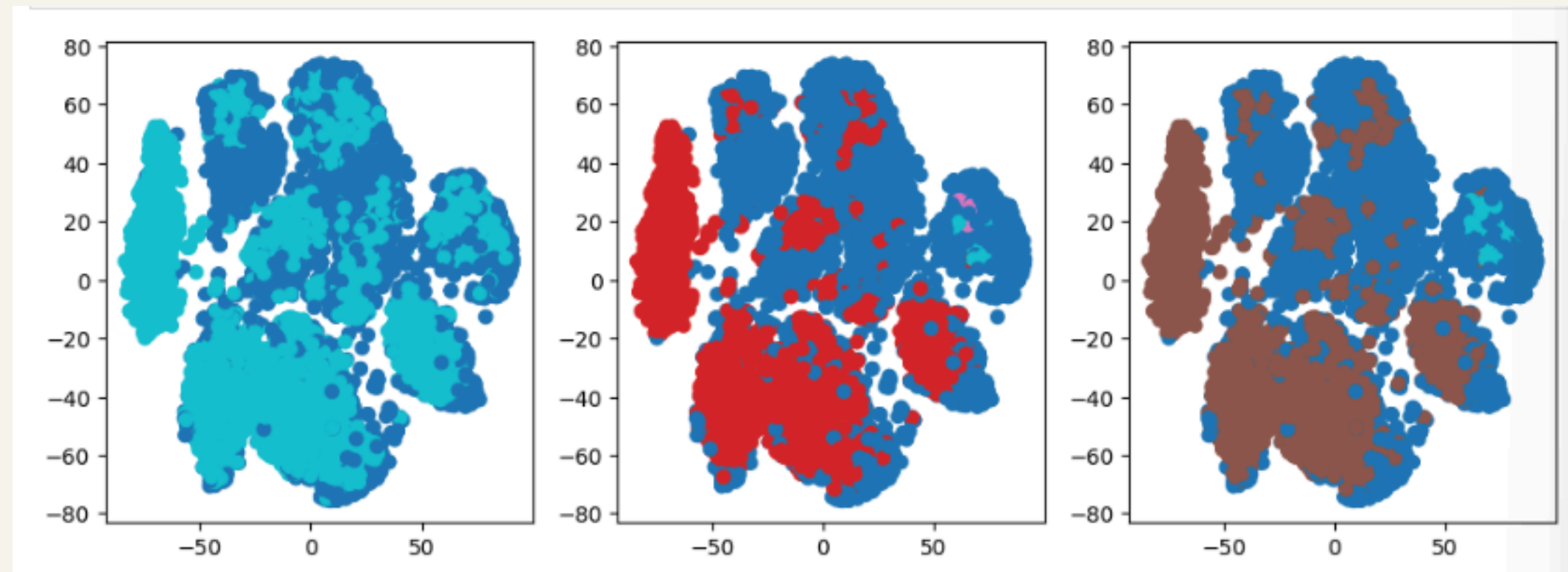
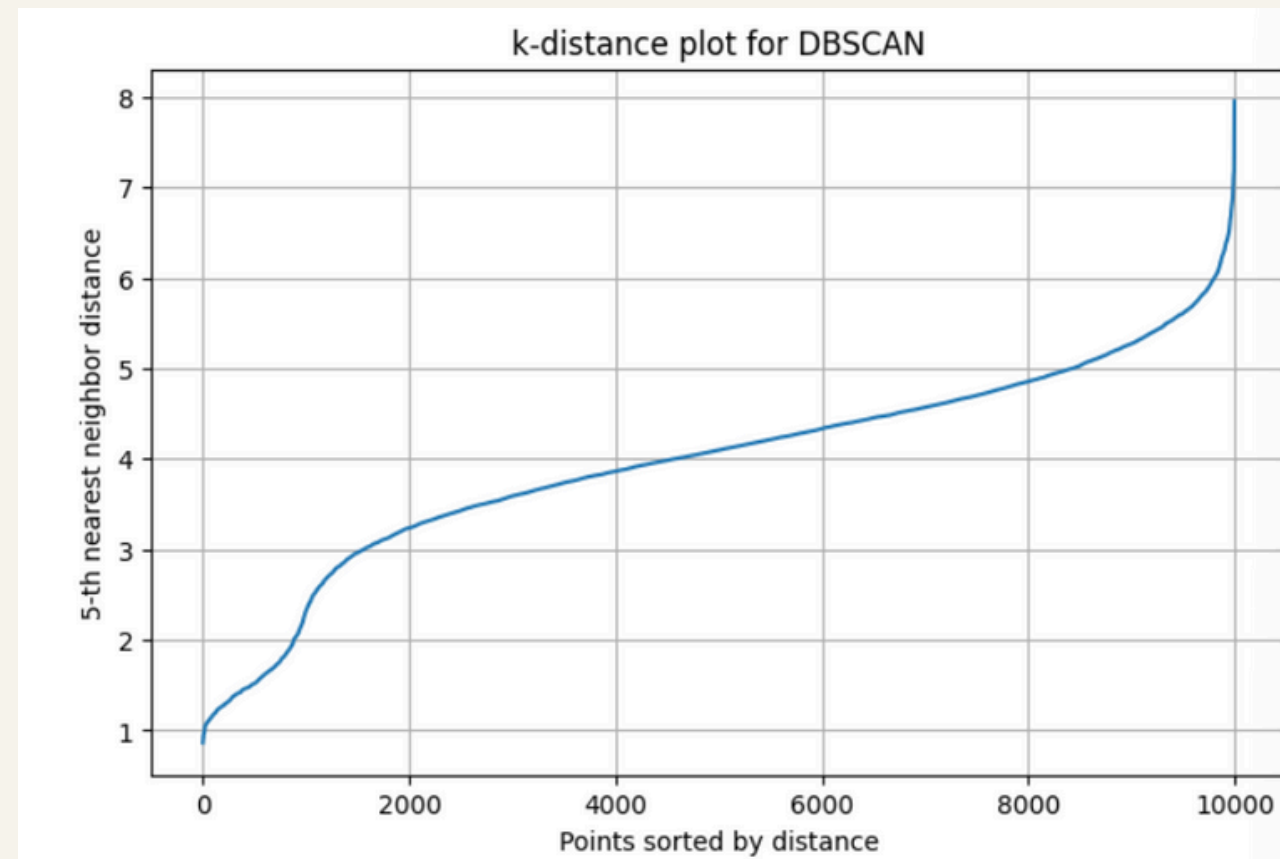


Agglomerative clustering moderately captures the structure of the MNIST dataset. While cluster compactness is low, the ARI and NMI values indicate a reasonable level of label consistency, making the method somewhat effective for unsupervised digit grouping.



Linkage=ward, Affinity=euclidean, Silhouette=0.053
Linkage=complete, Affinity=euclidean, Silhouette=0.037
Linkage=complete, Affinity=manhattan, Silhouette=0.012
Linkage=complete, Affinity=cosine, Silhouette=0.028
Linkage=average, Affinity=euclidean, Silhouette=0.047
Linkage=average, Affinity=manhattan, Silhouette=-0.005
Linkage=average, Affinity=cosine, Silhouette=0.057

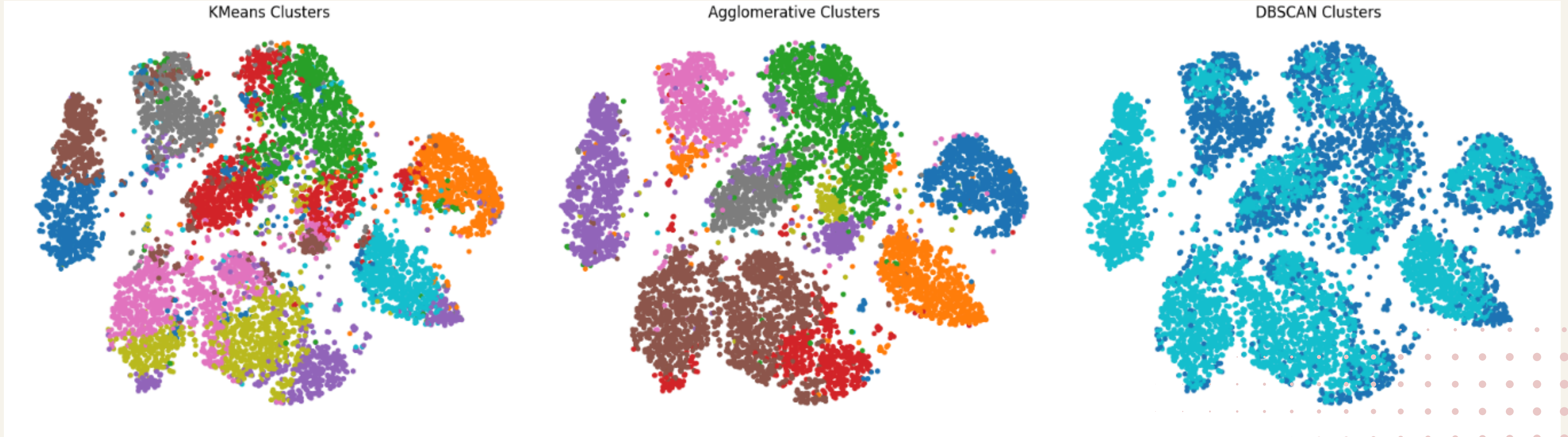
DBSCAN



```
sil_score: -0.025284382
ari_score: 0.0021887857167143194
nmi_score: 0.019683488304156695
```

DBSCAN was applied on PCA-reduced MNIST data to find digit clusters. Despite parameter tuning, it formed only 2–3 broad clusters with high noise. Visualizations using t-SNE showed overlapping digits. Due to uneven density and high dimensionality, DBSCAN is unsuitable for MNIST

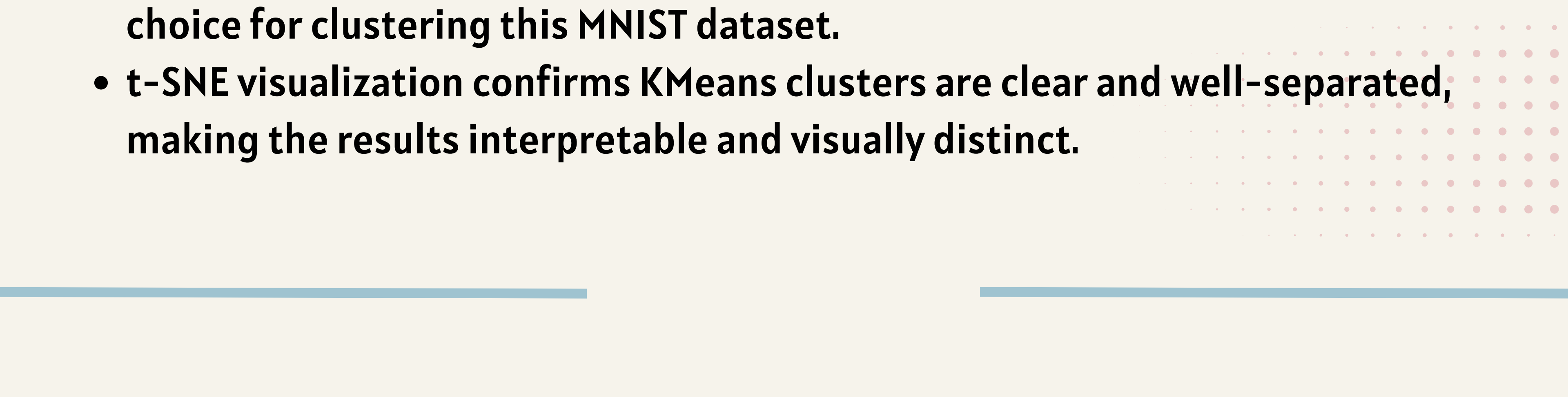
CHOOSING BEST CLUSTERING



	Clustering	Silhouette Score	ARI	NMI
0	KMeans	0.081542	0.361221	0.484099
1	Agglomerative	0.060073	0.507687	0.646572
2	DBSCAN	0.059225	0.037998	0.089142



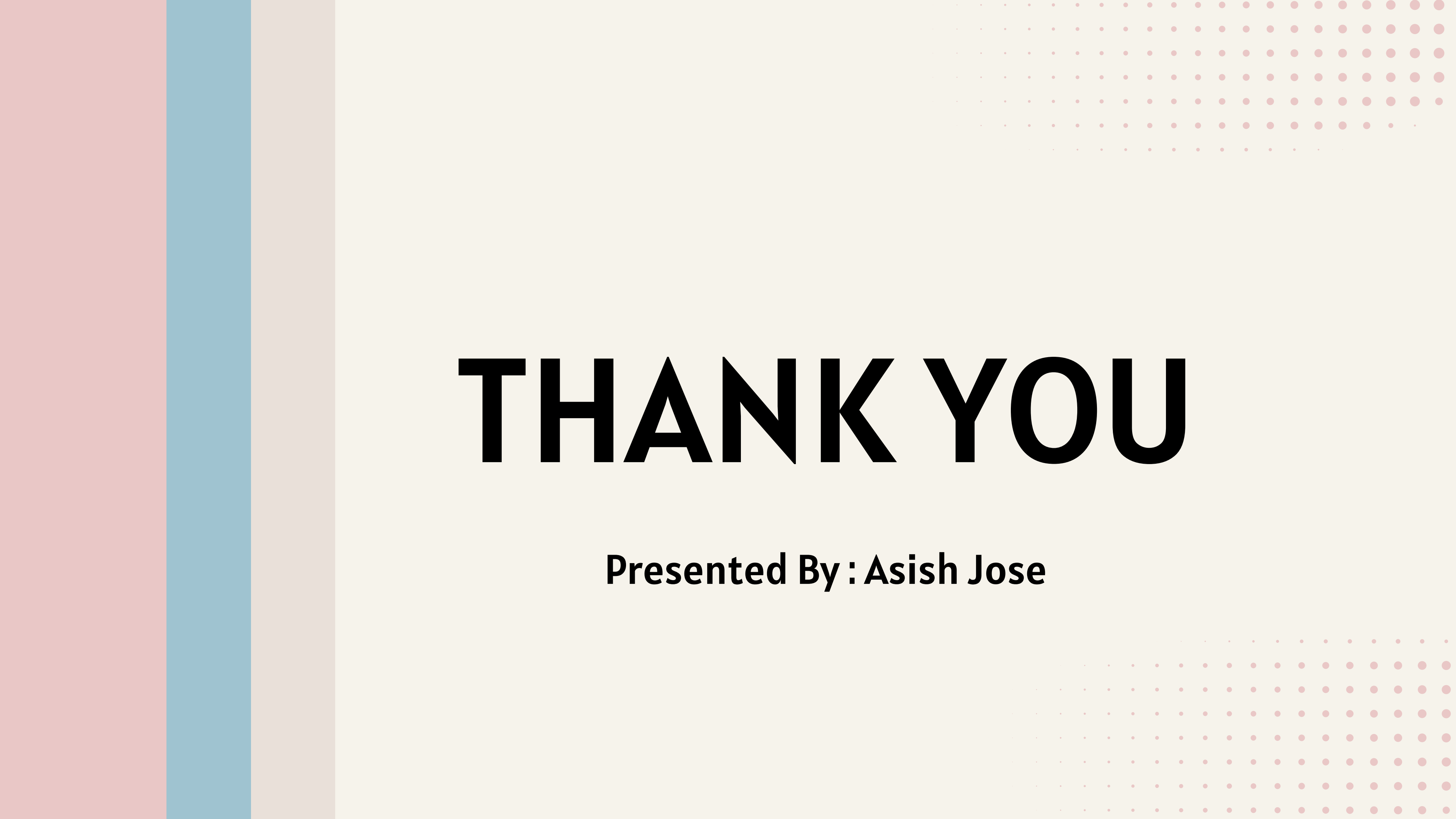
CONCLUSION

- **KMeans shows the highest Silhouette Score, indicating slightly better cluster cohesion.**
 - **Agglomerative Clustering achieves higher ARI and NMI, meaning its clusters align more closely with the true labels.**
 - **Considering both cohesion and label agreement, KMeans is a balanced choice for clustering this MNIST dataset.**
 - **t-SNE visualization confirms KMeans clusters are clear and well-separated, making the results interpretable and visually distinct.**
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FUTURE SCOPE

- **Use deep learning (autoencoders) to extract better features.**
 - **Enable accurate clustering of all 10 digits.**
 - **Improve visualization and interpretability of clusters.**
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The background features three vertical stripes on the left: a wide pink stripe, a narrower blue stripe, and a medium-width beige stripe. The right side of the image is a light cream color, decorated with two rectangular areas of a pink dot pattern. One area is in the top right corner, and the other is in the bottom right corner. Both dot patterns consist of small circles that gradually fade out towards the right edge.

THANK YOU

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